Rudraansh Patel

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Education

Maulana Azad National Institute of Technology (MANIT) — Bhopal, MP
B.Tech in Civil Engineering | CGPA: 6.76 — *2023 – 2027*

Indian Institute of Technology Madras (IIT Madras) — Online
BS in Data Science and Applications – Foundation Level — *2024*

Jawahar Navodaya Vidyalaya (JNV) — Amarkantak, MP
Class XII (CBSE): 97.8% — *2021*
Class X (CBSE): 93.2% — *2019*

Technical Skills

Languages: Python, C++, SQL, JavaScript
Backend & Databases: FastAPI, REST APIs, PostgreSQL, MySQL, SQLite, Redis
AI/ML: Retrieval-Augmented Generation (RAG), FAISS, Sentence-Transformers, Llama 3
Tools & Testing: Git, GitHub, GitHub Actions, Docker, pytest, VS Code
CS Fundamentals: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks

Projects

Shortie: High-Throughput URL Shortener | *Python, FastAPI, PostgreSQL, Redis, Docker* — 2026

- Built a URL shortener with REST APIs for creation, redirects and analytics, using base62-encoded IDs for collision-free codes.
- Added a Redis read-through cache for hot links, cutting p95 redirect latency from 38ms to 12ms.
- Added batched Redis click counters flushed to PostgreSQL, per-IP rate limiting, Docker Compose and pytest tests.

LeetCode Problem Finder | *JavaScript, GitHub Actions, GitHub Pages* — 2026

- Built redirects on rudraanshpatel.in where `/<number>` opens that LeetCode problem, using the 404 page as a client-side router.
- Wrote a scheduled GitHub Actions job that rebuilds a JSON index of 4,000+ problems server-side, avoiding browser CORS blocks.
- Reused the index in a Google Apps Script tracker, using CacheService to stay within daily fetch quotas.

ClauseWise: RAG Assistant for Indian Standard Codes | *Python, FastAPI, FAISS, BM25* — In Progress

- Building a QA assistant over Indian Standard codes, starting with IS 456:2000, with clause citations.
- Designing clause-aware chunking and cross-reference resolution to pull in related clauses and tables.
- Combining BM25 and embeddings for exact-term queries, benchmarked on coursework questions.

Leadership & Achievements

- **Lead Organizer, SIH Internal Hackathon @ MANIT (2024 & 2025):** Sole lead for two years running; scaled participation from **87 to 172 teams (98% growth)** and Grand Finale qualifications from **2 to 7 teams** through structured mentorship rounds.
- **Goldman Sachs India Hackathon (GSIH) 2026:** Ranked **1602 / 15,953 (top 10%)** in a 12-hour individual coding contest, with perfect scores on two algorithmic problems.
- Solved **400+ DSA problems** across LeetCode, Codeforces and CodeChef.
- **Vigyan Manthan Yatra (MP Government):** Ranked in the **top 100 statewide** and awarded a state scholarship, from 625 students selected across all 51 districts of Madhya Pradesh.